

Supplementary Materials for Real-Time Multimodal Activity-Aware Error Detection in Robot-Assisted Surgery

Seyed Hamid Reza Roodabeh^{*1}, Zongyu Li^{*1}, and Homa Alemzadeh¹

APPENDIX A: INSTRUMENT-OBJECT INTERACTIONS AND TEXTUAL PROMPTS

To construct the instrument–object interaction prompts and the interaction labels used for activity-aware visual encoder pretraining, we map surgical context states to structured action triplets. Let the context state at time t be

$$\mathbf{s}_t = [S_{1,t}, S_{2,t}, S_{3,t}, S_{4,t}, S_{5,t}], \quad (1)$$

where S_1 and S_2 denote the object held by and contacted by the left grasper, respectively, and S_3 and S_4 denote the object held by and contacted by the right grasper, respectively. The first four variables use the object coding defined in the main manuscript: 0 = nothing, 1 = ball/block, 2 = needle, 3 = thread, 4 = fabric/tissue, 5 = ring, and 6 = other. The fifth variable, S_5 , is task specific and represents the state of progress within a trial. For the suturing context shown in Fig. 1(c) of the main manuscript, S_5 represents the needle state, where 0, 1, and 2 denote not touching, touching, and in the fabric/tissue, respectively.

We represent each low-level interaction using an action-triplet format

$$\langle \text{instrument}, \text{verb}, \text{target} \rangle, \quad (2)$$

which provides a structured representation of the instrument, the interaction type, and the target object, inspired by [1]. Table I shows the mapping from context-state patterns to the triplet labels used in our prompt generation and activity-aware pretraining. In the pattern column, an asterisk indicates a wildcard that can take any valid context-state value. Thus, a pattern matches a context state if all non-wildcard entries match the corresponding entries of $\mathbf{s}_t$.

Because wildcard patterns are used, one triplet can correspond to multiple context states. For example, the triplet $\langle \text{left}, \text{hold}, \text{needle} \rangle$ matches any context state of the form $2****$, where the first state variable indicates that the left grasper is holding the needle and the remaining state variables may take any value.

Conversely, a single context state can match more than one triplet. For example, the context-state pattern $2***1$ is consistent with both $\langle \text{left}, \text{hold}, \text{needle} \rangle$ and $\langle \text{left}, \text{hold-needle}, \text{touch-fabric} \rangle$. When a single interaction label or prompt phrase is required, we use the more specific composite triplet, because it encodes both the grasper–needle

TABLE I
MAPPING FROM SURGICAL CONTEXT-STATE PATTERNS TO
INSTRUMENT–OBJECT INTERACTION TRIPLETS. THE CONTEXT-STATE
PATTERN IS ORDERED AS $[S_1, S_2, S_3, S_4, S_5]$. AN ASTERISK DENOTES A
WILDCARD.

Index	Instrument	Verb	Target	Context-state pattern
0	left	hold	needle	2****
1	left	hold	thread	3****
2	left	contact	needle	*2***
3	left	contact	thread	*3***
4	left	hold-needle	nottouch-fabric	2***0
5	left	hold-needle	touch-fabric	2***1
6	left	hold-needle	in-fabric	2***2
7	right	hold	needle	**2**
8	right	hold	thread	**3**
9	right	contact	needle	***2*
10	right	contact	thread	***3*
11	right	hold-needle	nottouch-fabric	**2*0
12	right	hold-needle	touch-fabric	**2*1
13	right	hold-needle	in-fabric	**2*2
14	left	null	null	00000 (left exists)
15	right	null	null	00000 (right exists)
16	null	null	null	00000 (both not exist)

grip and the concurrent needle–fabric interaction. Therefore, $2***1$ is rendered as the left grasper holding the needle while the needle touches the fabric/tissue.

The resulting triplets are converted into concise natural-language phrases for the interaction prompts. For example, $\langle \text{left}, \text{hold}, \text{needle} \rangle$ is rendered as “holding the needle with the left grasper”, while $\langle \text{right}, \text{hold-needle}, \text{in-fabric} \rangle$ is rendered as “holding the needle with the right grasper while the needle is in the fabric/tissue”. These interaction phrases are used directly in the Instrument–Object Interaction prompts and are combined with gesture goals and error descriptions in the Interaction+Error prompts described in the main manuscript.

APPENDIX B: TEXTUAL PROMPTS

This section provides the exhaustive lists of textual prompts utilized for the activity-aware feature contextualization across both the SAR-RARP50 and JIGSAWS datasets. As a reference, the definitions of gestures for both JIGSAWS and SAR-RARP50 datasets are provided in Table II and Table III, respectively. The total number of prompts for each prompt set and each dataset is presented in Table IV.

A. SAR-RARP50 Prompts

1) Error Prompts:

- A surgeon is making multiple attempts.
- A surgeon is dropping or slipping the needle in the tissue.
- A surgeon is working with an instrument out of view.
- A surgeon is working with the needle out of view.

^{*}These authors have contributed equally.

¹The authors are with the Electrical and Computer Engineering Department, School of Engineering and Applied Science, University of Virginia, Charlottesville, VA, United States of America {ydg9ag, z17qgw, ha4d}@virginia.edu

TABLE II
GESTURE DEFINITIONS FOR THE JIGSAWS DATASET

Gesture ID	Definition
G1	Reaching for the needle with right hand
G2	Positioning the tip of the needle
G3	Pushing needle through the tissue
G4	Transferring needle from left to right
G5	Moving to center of workspace with needle in grip
G6	Pulling suture with left hand
G7	Pulling suture with right hand
G8	Orienting needle
G9	Using right hand to help tighten suture
G10	Loosening more suture
G11	Dropping suture and moving to end points
G12	Reaching for needle with left hand
G13	Making C loop around right hand
G14	Reaching for suture with right hand
G15	Pulling suture with both hands

TABLE III
GESTURE DEFINITIONS FOR THE SAR-RARP50 DATASET

Gesture ID	Definition
G0	Other
G1	Picking-up the needle
G2	Positioning the needle tip
G3	Pushing the needle through the tissue
G4	Pulling the needle out of the tissue
G5	Tying a knot
G6	Cutting the suture
G7	Returning/dropping the needle

TABLE IV
PROMPT COUNTS PER CATEGORY AND DATASET.

Prompt Type	JIGSAWS	SAR-RARP50
Gesture	15	8
Interaction	26	26
Error	7	24
Gesture + Error	34	44
Interaction + Error	34	44

- A surgeon is causing tissue damage, including poor or error in tissue stabilisation.
- A surgeon is grasping the needle at a non-perpendicular angle.
- A surgeon is holding the needle incorrectly along its length.
- A surgeon is applying excessive force.
- A surgeon is not following the curve of the needle.
- A surgeon is inserting the needle at a non-perpendicular angle.
- A surgeon is grasping at the needle tip.
- A surgeon is loosening the suture.
- A surgeon is making the thread caught in the instrument.
- A surgeon is tying a knot that is not square.
- A surgeon is making an inadequate number of throws.
- A surgeon is pulling the suture through tissue before tying the knot.
- A surgeon is placing needle drives too close or too far apart.
- A surgeon is not pulling the suture through between needle drives.
- A surgeon is entangling the suture.
- A surgeon is fraying the suture.
- A surgeon is snapping the suture.
- A surgeon is disposing of the needle in a dangerous, poor or incorrect manner.
- A surgeon is working with a blurred or poor view due to incorrect camera control.
- A surgeon is exercising incorrect or poor instrument control, clashing, using a 3rd arm, or non-dominant use.
- A surgeon is executing the task perfectly.

2) Gesture Prompts:

- **G0:** A surgeon is performing another action.
- **G1:** A surgeon is picking up the needle.
- **G2:** A surgeon is positioning the needle tip.
- **G3:** A surgeon is pushing the needle through the tissue.
- **G4:** A surgeon is pulling the needle out of the tissue.
- **G5:** A surgeon is tying a knot.
- **G6:** A surgeon is cutting the suture.
- **G7:** A surgeon is returning/dropping the needle.

3) Interaction Prompts:

- A surgeon is holding the needle with the left grasper
- A surgeon is holding the suture thread with the left grasper
- A surgeon is holding a ring with the left grasper
- A surgeon is holding the needle with the left grasper in contact
- A surgeon is holding the suture thread with the left grasper in contact
- A surgeon is holding a ring with the left grasper in contact
- A surgeon is holding the needle with the left grasper, not touching the fabric
- A surgeon is holding the needle with the left grasper, touching the fabric
- A surgeon is holding the needle with the left grasper, inserting it into the fabric
- A surgeon is holding the needle with the right grasper
- A surgeon is holding the suture thread with the right grasper
- A surgeon is holding a ring with the right grasper
- A surgeon is holding the needle with the right grasper in contact
- A surgeon is holding the suture thread with the right grasper in contact
- A surgeon is holding a ring with the right grasper in contact
- A surgeon is holding the needle with the right grasper, not touching the fabric
- A surgeon is holding the needle with the right grasper, touching the fabric
- A surgeon is holding the needle with the right grasper, inserting it into the fabric
- A surgeon is idle with the left grasper (not interacting with the needle or thread)
- A surgeon is idle with the right grasper (not interacting with the needle or thread)

4) Gesture + Error Prompts:

- **G0:** A surgeon is performing another action but makes multiple attempts / with an instrument out of view / with poor instrument control or clashing / with a blurred or poor view due to incorrect camera control / achieving perfect execution.
- **G1:** A surgeon is picking up the needle but makes multiple attempts / drops or slips the needle in the tissue / working with the instrument out of view / grasps the needle at a non-perpendicular angle / holding the needle incorrectly along its length / grasps at the needle tip / achieving perfect execution.
- **G2:** A surgeon is positioning the needle tip but makes multiple attempts / working with the needle out of view / grasps the needle at a non-perpendicular angle / holding the needle incorrectly along its length / applying excessive force / achieving perfect execution.
- **G3:** A surgeon is pushing the needle through the tissue but makes multiple attempts / working with the needle out of view / causing tissue damage / not following the curve of the needle / inserting the needle at a non-perpendicular angle / applying excessive force / achieving perfect execution.
- **G4:** A surgeon is pulling the needle out of the tissue but makes multiple attempts / working with the needle out of view / causing tissue damage / fraying the suture / snapping the suture / achieving perfect execution.
- **G5:** A surgeon is tying a knot that is not square / making an inadequate number of throws / entangles the suture / loosens the suture / makes multiple attempts / the thread is caught in the instrument / with poor instrument control / achieving perfect execution.
- **G6:** A surgeon is cutting the suture but frays the suture / snaps

the suture / achieving perfect execution.

- **G7:** A surgeon is returning the needle but disposes of it in a dangerous, poor, or incorrect manner / achieving perfect execution.

5) Interaction + Error Prompts:

- **G0:** A surgeon is performing another action but makes multiple attempts / with an instrument out of view / with poor instrument control or clashing / with a blurred or poor view due to incorrect camera control / achieving perfect execution.
- **G1:** A surgeon is not holding anything then holding the needle with right grasper, needle in the fabric but makes multiple attempts / drops or slips the needle in the tissue / working with the instrument out of view / grasps the needle at a non-perpendicular angle / holding the needle incorrectly along its length / grasps at the needle tip / achieving perfect execution.
- **G2:** A surgeon is holding the needle with grasper, then needle touches the tissue but makes multiple attempts / working with the needle out of view / grasps the needle at a non-perpendicular angle / holding the needle incorrectly along its length / applying excessive force / achieving perfect execution.
- **G3:** A surgeon is holding the needle with grasper, then needle in the tissue but makes multiple attempts / working with the needle out of view / causing tissue damage / not following the curve of the needle / inserting the needle at a non-perpendicular angle / applying excessive force / achieving perfect execution.
- **G4:** A surgeon is holding the needle with grasper, then needle not touches the tissue but makes multiple attempts / working with the needle out of view / causing tissue damage / fraying the suture / snapping the suture / achieving perfect execution.
- **G5:** A surgeon is holding the thread with grasper, forming a loop, maintaining tension, and cinching the knot onto the tissue but the knot is not square / makes an inadequate number of throws / entangles the suture / loosens the suture / makes multiple attempts / the thread is caught in the instrument / with poor instrument control / achieving perfect execution.
- **G6:** A surgeon is holding the scissors, then the scissors are closed to sever the suture but frays the suture / snaps the suture / achieving perfect execution.
- **G7:** A surgeon is holding the needle with an instrument, moving it toward a neutral area or dropping it safely onto a surface but disposes of it in a dangerous, poor, or incorrect manner.
- **G7 (Alt):** A surgeon is holding the scissors, then the scissors are closed to sever the suture, achieving perfect execution.

B. JIGSAWS Prompts

1) Error Prompts:

- The surgeon is making multiple attempts.
- The surgeon is dropping the needle.
- The surgeon is losing the needle out of view.
- The surgeon is executing the task perfectly.
- The surgeon is not moving the needle along the needle curve.
- The surgeon is grasping the needle too far or too short.
- The surgeon is holding the needle not perpendicular to the tissue.

2) Gesture Prompts:

- A surgeon is performing another action
- A surgeon is reaching for needle with right hand
- A surgeon is positioning needle
- A surgeon is pushing needle through tissue
- A surgeon is transferring needle from left to right
- A surgeon is moving to center with needle in grip
- A surgeon is pulling suture with left hand
- A surgeon is orienting needle
- A surgeon is using right hand to help tighten suture

3) Interaction Prompts:

- A surgeon is holding the needle with the left grasper
- A surgeon is holding the suture thread with the left grasper
- A surgeon is holding a ring with the left grasper

- A surgeon is holding the needle with the left grasper in contact
- A surgeon is holding the suture thread with the left grasper in contact
- A surgeon is holding a ring with the left grasper in contact
- A surgeon is holding the needle with the left grasper, not touching the fabric
- A surgeon is holding the needle with the left grasper, touching the fabric
- A surgeon is holding the needle with the left grasper, inserting it into the fabric
- A surgeon is holding the needle with the left grasper, not touching a ring
- A surgeon is holding the needle with the left grasper, touching a ring
- A surgeon is holding the needle with the left grasper, inserting it into a ring
- A surgeon is holding the needle with the right grasper
- A surgeon is holding the suture thread with the right grasper
- A surgeon is holding a ring with the right grasper
- A surgeon is holding the needle with the right grasper in contact
- A surgeon is holding the suture thread with the right grasper in contact
- A surgeon is holding a ring with the right grasper in contact
- A surgeon is holding the needle with the right grasper, not touching the fabric
- A surgeon is holding the needle with the right grasper, touching the fabric
- A surgeon is holding the needle with the right grasper, inserting it into the fabric
- A surgeon is holding the needle with the right grasper, not touching a ring
- A surgeon is holding the needle with the right grasper, touching a ring
- A surgeon is holding the needle with the right grasper, inserting it into a ring
- A surgeon is idle with the left grasper (not interacting with the needle or thread)
- A surgeon is idle with the right grasper (not interacting with the needle or thread)

4) Gesture + Error Prompts:

- A surgeon is reaching for the needle with the right hand and makes multiple attempts / but the needle drops / but the needle is out of view / achieving perfect execution.
- A surgeon is positioning the needle but makes multiple attempts / but the needle drops / but the needle is out of view / achieving perfect execution.
- A surgeon is pushing the needle through the tissue, but the needle is not moving along the needle curve / but makes multiple attempts / but the needle drops / but the needle is out of view / achieving perfect execution.
- A surgeon is transferring the needle from left to right but makes multiple attempts / but grasps too far/too short on the needle / but the needle drops / but the needle is out of view / achieving perfect execution.
- A surgeon is moving to the center with the needle in grip, but the needle drops / but the needle is out of view / achieving perfect execution.
- A surgeon is pulling the suture with the left hand but makes multiple attempts / but the needle drops / but the needle is out of view / achieving perfect execution.
- A surgeon is orienting the needle but makes multiple attempts / but it is not perpendicular to the tissue / but the needle drops / but the needle is out of view / achieving perfect execution.
- A surgeon is using the right hand to help tighten the suture but makes multiple attempts / but the needle drops / but the needle is out of view / achieving perfect execution.

5) Interaction + Error Prompts:

- A surgeon is holding the needle with the right grasper but makes

multiple attempts / but the needle drops / but the needle is out of view / achieving perfect execution.

- A surgeon is holding the needle with the right grasper to touch the fabric or ring but makes multiple attempts / but the needle drops / but the needle is out of view / achieving perfect execution.
- A surgeon is holding the needle with the right grasper to push the needle into the fabric or ring but the needle is not moving along the needle curve / but makes multiple attempts / but the needle drops / but the needle is out of view / achieving perfect execution.
- A surgeon is holding the needle with the right grasper from the left grasper but makes multiple attempts / but grasps too far or too short on the needle / but the needle drops / but the needle is out of view / achieving perfect execution.
- A surgeon is holding the needle with the right grasper to contact the left grasper but the needle drops / but the needle is out of view / achieving perfect execution.
- A surgeon is holding the needle with the right grasper to release contact with the fabric or ring but makes multiple attempts / but the needle drops / but the needle is out of view / achieving perfect execution.
- A surgeon is holding the needle with the right grasper while switching between the left grasper but makes multiple attempts / but it is not perpendicular to the grasper / but the needle drops / but the needle is out of view / achieving perfect execution.
- A surgeon is holding the needle with the left grasper to let the right grasper contact the thread but makes multiple attempts / but the needle drops / but the needle is out of view / achieving perfect execution.

APPENDIX C: VISION ENCODER ABLATION

We have conducted an ablation study on the more surgically relevant SAR-RARP50 dataset to evaluate alternative general-domain vision encoders. We used the Activity Prompting model which uses only images as windowed inputs to further emphasize the effect of the visual encoder module. Our results, summarized in Table V, show that while more expressive models demonstrate improved performance on certain prompts, the pre-trained ResNet50 maintains the most balanced and competitive overall performance. Specifically, our models with the ResNet50 encoder achieve an end-to-end detection latency of around 35 ms, while the larger Dinov2-Large encoder alone yields latencies of up to 140 ms on consumer hardware. This observation further demonstrates the feasibility of precise detection with frequent, near 30 Hz model executions using the ResNet50 encoder.

APPENDIX D: DATASET DISTRIBUTIONS

While the LOSO folds from the JIGSAWS dataset, shown in Table VI, have a large variation in the total number of windows and ratio of erroneous windows, the distribution in SAR-RARP50 dataset, as shown in Table VII, maintains a more balanced ratio between nominal and erroneous windows across both the training and test splits.

APPENDIX E: CROSS VALIDATION RESULTS

We present comprehensive performance results using Leave-One-User-Out (LOUO) setup, in addition to the LOSO method discussed in the main manuscript. As detailed in Table VIII and consistent with previous studies, model performance is lower in the LOUO setting compared to LOSO. We attribute this performance gap largely to inter-operator variations in surgical technique and the subsequent visual/kinematics manifestations.

APPENDIX F: COMPLEXITY PROFILE OF MODELS

We conducted a thorough complexity profile analysis of our models and selected baseline methods. This evaluation was performed using an NVIDIA RTX 3090 Ti GPU (with 24 GB of memory), an Intel Core-i7-13700K CPU, limited to 8 cores, and 16 GB of RAM. We utilized a standard input window of 10 samples (2 seconds). We measured key complexity and inference metrics, such as the number of parameters for the vision encoder and error detection models, FLOPs per window, streaming latency, and end-to-end latency for each model configuration, as shown in Table IX. Here, streaming latency is the time taken to prepare and preprocess a model's input window prior to passing it into the relevant encoders. For vision encoders, it includes converting images from in-memory RGB representations into PIL format, resizing, center-cropping and normalization, and conversion to Torch tensors. For text encoders, it involves text preprocessing and tokenization, while for the overall model, it includes boolean mask generation and kinematics normalization. All of these pre-processing steps involve CPU-heavy operations. Our proposed architectures remain highly competitive in inference efficiency, achieving a model inference latency of ~ 1.49 – 1.64 ms per window and an end-to-end latency of around 35 ms. Given recent advancements in GPU-enabled edge platforms (e.g., NVIDIA Jetson and Thor), these metrics confirm that our framework is suitable for real-time, low-latency error detection at the surgical edge. Furthermore, no model with a ResNet50 encoder utilized more than 8 GB of GPU memory during training or inference.

REFERENCES

- [1] C. I. Nwoye, C. Gonzalez, T. Yu, P. Mascagni, D. Mutter, J. Marescaux, and N. Padoy, "Recognition of instrument-tissue interactions in endoscopic videos via action triplets," in *Medical Image Computing and Computer Assisted Intervention-MICCAI 2020: 23rd International Conference, Lima, Peru, October 4–8, 2020, Proceedings, Part III* 23. Springer, 2020, pp. 364–374.

TABLE V
ABLATION STUDY ON VISION ENCODERS (ACTIVITY PROMPTING)

Vision Encoder	Prompt Type	F1	Accuracy	Jaccard	Latency (ms)
ResNet-50	Interaction	0.713 $\pm$ 0.052	0.564 $\pm$ 0.041	0.554 $\pm$ 0.060	35.46 $\pm$ 0.63
ResNet-50	Error	0.708 $\pm$ 0.048	0.566 $\pm$ 0.038	0.548 $\pm$ 0.047	36.64 $\pm$ 0.32
ResNet-50	Gesture	0.735 $\pm$ 0.033	0.613 $\pm$ 0.042	0.594 $\pm$ 0.051	35.63 $\pm$ 0.04
ResNet-50	Gesture + Error	0.712 $\pm$ 0.028	0.578 $\pm$ 0.036	0.553 $\pm$ 0.014	35.53 $\pm$ 0.07
ResNet-50	Interaction + Error	0.747 $\pm$0.021	0.624 $\pm$0.017	0.596 $\pm$0.039	35.32 $\pm$0.59
CLIP-ViT-B/32	Interaction	0.730 $\pm$ 0.085	0.598 $\pm$ 0.072	0.575 $\pm$ 0.091	36.19 $\pm$ 1.41
CLIP-ViT-B/32	Error	0.699 $\pm$ 0.093	0.577 $\pm$ 0.088	0.538 $\pm$ 0.079	35.40 $\pm$ 0.89
CLIP-ViT-B/32	Gesture	0.733 $\pm$ 0.061	0.583 $\pm$ 0.068	0.578 $\pm$ 0.075	35.97 $\pm$ 0.61
CLIP-ViT-B/32	Gesture + Error	0.736 $\pm$ 0.055	0.590 $\pm$ 0.064	0.583 $\pm$ 0.042	35.44 $\pm$ 1.08
CLIP-ViT-B/32	Interaction + Error	0.743 $\pm$0.041	0.609 $\pm$0.035	0.591 $\pm$0.058	36.00 $\pm$1.74
DINOv2-Large	Interaction	0.731 $\pm$ 0.042	0.579 $\pm$ 0.038	0.576 $\pm$ 0.052	138.76 $\pm$ 0.70
DINOv2-Large	Error	0.728 $\pm$ 0.065	0.574 $\pm$ 0.021	0.573 $\pm$ 0.039	138.22 $\pm$ 2.12
DINOv2-Large	Gesture	0.731 $\pm$ 0.029	0.618 $\pm$ 0.025	0.589 $\pm$ 0.014	137.22 $\pm$ 0.40
DINOv2-Large	Gesture + Error	0.731 $\pm$ 0.049	0.579 $\pm$ 0.014	0.576 $\pm$ 0.018	138.56 $\pm$ 0.70
DINOv2-Large	Interaction + Error	0.747 $\pm$0.015	0.620 $\pm$0.018	0.596 $\pm$0.011	139.65 $\pm$0.68

TABLE VI
WINDOW-LEVEL DISTRIBUTION ACROSS LOSO SETTINGS FOR THE JIGSAWS DATASET.

LOSO Fold	Total Windows	Erroneous Windows
Trial 1 out	1,076	764 (71%)
Trial 2 out	775	466 (60%)
Trial 3 out	886	500 (56%)
Trial 4 out	868	448 (52%)
Trial 5 out	640	308 (48%)

TABLE VII
WINDOW AND ERROR DISTRIBUTION ACROSS DATASET SPLITS FOR THE SAR-RARP50 DATASET

Set	Total Windows	Erroneous Windows
Train	10,614	4,727 (44.5%)
Test	1,599	653 (41.0%)

TABLE VIII
LOUO AND LOSO EVALUATION METRICS FOR ACTIVITY PROMPTING AND KINEMATICS FUSION

Prompt Type	LOUO			LOSO		
	F1	Accuracy	Jaccard	F1	Accuracy	Jaccard
<i>Activity Prompting</i>						
Interaction	0.657 $\pm$ 0.112	0.553 $\pm$ 0.139	0.509 $\pm$ 0.127	0.724 $\pm$ 0.052	0.664 $\pm$ 0.031	0.570 $\pm$ 0.066
Error	0.668 $\pm$ 0.105	0.584 $\pm$ 0.103	0.510 $\pm$ 0.121	0.733 $\pm$ 0.051	0.675 $\pm$ 0.033	0.598 $\pm$ 0.065
Gesture	0.646 $\pm$ 0.106	0.581 $\pm$ 0.081	0.519 $\pm$ 0.124	0.712 $\pm$ 0.056	0.655 $\pm$ 0.040	0.555 $\pm$ 0.072
Gesture + Error	0.670 $\pm$ 0.114	0.544 $\pm$ 0.121	0.513 $\pm$ 0.132	0.745 $\pm$ 0.047	0.692 $\pm$ 0.032	0.588 $\pm$ 0.059
Interaction + Error	0.663 $\pm$ 0.123	0.508 $\pm$ 0.138	0.507 $\pm$ 0.140	0.730 $\pm$ 0.055	0.671 $\pm$ 0.035	0.553 $\pm$ 0.068
<i>Activity Kinematics Fusion</i>						
Interaction	0.653 $\pm$ 0.124	0.512 $\pm$ 0.142	0.507 $\pm$ 0.140	0.724 $\pm$ 0.054	0.687 $\pm$ 0.025	0.570 $\pm$ 0.068
Error	0.664 $\pm$ 0.106	0.533 $\pm$ 0.126	0.505 $\pm$ 0.119	0.745 $\pm$ 0.048	0.702 $\pm$ 0.062	0.591 $\pm$ 0.061
Gesture	0.653 $\pm$ 0.123	0.547 $\pm$ 0.140	0.507 $\pm$ 0.140	0.729 $\pm$ 0.047	0.666 $\pm$ 0.040	0.565 $\pm$ 0.060
Gesture + Error	0.684 $\pm$ 0.110	0.561 $\pm$ 0.109	0.517 $\pm$ 0.128	0.760 $\pm$ 0.044	0.717 $\pm$ 0.027	0.634 $\pm$ 0.056
Interaction + Error	0.695 $\pm$ 0.109	0.583 $\pm$ 0.094	0.518 $\pm$ 0.127	0.749 $\pm$ 0.043	0.699 $\pm$ 0.027	0.606 $\pm$ 0.054

TABLE IX
MODEL COMPLEXITY AND LATENCY SUMMARY.

Model	Prompt Type	Encoder	Enc. Params	Model Params	FLOPs/ Window	Streaming Lat. (ms)	Encoder Lat. (ms)	Model Lat. (ms)	E2E Lat. (ms)
COG	Gesture	ResNet50	23.5M	1.8M	81.8B	27.35 $\pm$ 0.20	5.73 $\pm$ 0.01	17.06 $\pm$ 0.02	51.18 $\pm$ 0.58
SEDMamba	n/a	DINOV2-Giant	1.1B	290.0K	5.9T	21.81 $\pm$ 0.04	344.59 $\pm$ 0.10	0.7 $\pm$ 0.03	368.06 $\pm$ 0.52
Activity Prompting	Error	ResNet50	23.5M	6.2M	81.8B	27.27 $\pm$ 0.55	6.10 $\pm$ 0.12	1.49 $\pm$ 0.01	36.64 $\pm$ 0.32
Activity Kinematics Fusion	Gesture+Error	ResNet50	23.5M	9.7M	82.0B	26.54 $\pm$ 0.20	6.17 $\pm$ 0.19	1.54 $\pm$ 0.01	35.53 $\pm$ 0.07
Activity Prompting	Gesture	ResNet50	23.5M	6.4M	81.8B	26.88 $\pm$ 0.02	6.02 $\pm$ 0.02	1.50 $\pm$ 0.01	35.63 $\pm$ 0.04
Activity Prompting	Interaction+Error	ResNet50	23.5M	9.7M	82.0B	26.88 $\pm$ 0.02	6.16 $\pm$ 0.21	1.53 $\pm$ 0.01	35.32 $\pm$ 0.59
Activity Prompting	Interaction	ResNet50	23.5M	8.7M	81.9B	28.07 $\pm$ 1.63	6.17 $\pm$ 0.17	1.55 $\pm$ 0.00	35.46 $\pm$ 0.63
Activity Kinematics Fusion	Error	ResNet50	23.5M	6.5M	81.8B	26.90 $\pm$ 0.14	6.00 $\pm$ 0.03	1.58 $\pm$ 0.01	35.34 $\pm$ 0.21
Activity Kinematics Fusion	Gesture+Error	ResNet50	23.5M	10.0M	82.0B	26.63 $\pm$ 0.35	6.49 $\pm$ 0.76	1.63 $\pm$ 0.01	35.55 $\pm$ 0.03
Activity Kinematics Fusion	Gesture	ResNet50	23.5M	6.7M	81.8B	26.36 $\pm$ 0.03	6.08 $\pm$ 0.11	1.60 $\pm$ 0.00	35.81 $\pm$ 0.06
Activity Kinematics Fusion	Interaction+Error	ResNet50	23.5M	10.0M	82.0B	26.68 $\pm$ 0.02	6.19 $\pm$ 0.20	1.63 $\pm$ 0.01	35.90 $\pm$ 0.03
Activity Kinematics Fusion	Interaction	ResNet50	23.5M	9.0M	81.9B	26.90 $\pm$ 0.12	6.23 $\pm$ 0.17	1.64 $\pm$ 0.01	35.61 $\pm$ 0.10
Activity-Aware Embeddings	n/a	ResNet50	47M	42.0M	82.6B	27.08 $\pm$ 0.23	11.94 $\pm$ 0.06	1.36 $\pm$ 0.00	41.59 $\pm$ 0.68
Activity Prompting	Error	DINOV2-Large	304.4M	5.7M	1.5T	23.42 $\pm$ 0.14	112.52 $\pm$ 0.08	1.49 $\pm$ 0.01	138.22 $\pm$ 2.12
Activity Prompting	Gesture+Error	DINOV2-Large	304.4M	9.2M	1.5T	24.33 $\pm$ 0.64	112.71 $\pm$ 0.03	1.54 $\pm$ 0.01	138.56 $\pm$ 0.70
Activity Prompting	Gesture	DINOV2-Large	304.4M	5.9M	1.5T	23.01 $\pm$ 0.77	112.54 $\pm$ 0.01	1.49 $\pm$ 0.00	137.22 $\pm$ 0.40
Activity Prompting	Interaction+Error	DINOV2-Large	304.4M	9.2M	1.5T	23.43 $\pm$ 0.36	112.67 $\pm$ 0.08	1.53 $\pm$ 0.01	139.65 $\pm$ 0.68
Activity Prompting	Interaction	DINOV2-Large	304.4M	8.2M	1.5T	22.66 $\pm$ 0.39	112.63 $\pm$ 0.10	1.54 $\pm$ 0.00	138.76 $\pm$ 0.70
Activity Kinematics Fusion	Error	DINOV2-Large	304.4M	5.9M	1.5T	23.58 $\pm$ 1.19	112.57 $\pm$ 0.01	1.59 $\pm$ 0.00	138.64 $\pm$ 0.20
Activity Kinematics Fusion	Gesture+Error	DINOV2-Large	304.4M	9.5M	1.5T	24.07 $\pm$ 2.30	112.58 $\pm$ 0.04	1.63 $\pm$ 0.01	137.51 $\pm$ 0.32
Activity Kinematics Fusion	Gesture	DINOV2-Large	304.4M	6.2M	1.5T	22.84 $\pm$ 0.46	112.56 $\pm$ 0.01	1.59 $\pm$ 0.01	138.47 $\pm$ 0.95
Activity Kinematics Fusion	Interaction+Error	DINOV2-Large	304.4M	9.5M	1.5T	23.31 $\pm$ 1.07	112.71 $\pm$ 0.04	1.63 $\pm$ 0.00	137.88 $\pm$ 0.67
Activity Kinematics Fusion	Interaction	DINOV2-Large	304.4M	8.4M	1.5T	22.58 $\pm$ 0.01	112.66 $\pm$ 0.07	1.64 $\pm$ 0.00	138.41 $\pm$ 1.09
Activity-Aware Embeddings	n/a	DINOV2-Large	608.8M	3.2M	1.5T	23.99 $\pm$ 0.47	225.26 $\pm$ 0.22	0.93 $\pm$ 0.00	249.48 $\pm$ 1.12
Activity Prompting	Error	CLIP ViT-B/32	87.5M	5.5M	87.3B	22.65 $\pm$ 0.44	9.97 $\pm$ 0.01	1.57 $\pm$ 0.09	35.40 $\pm$ 0.89
Activity Prompting	Gesture+Error	CLIP ViT-B/32	87.5M	9.1M	87.5B	23.39 $\pm$ 1.08	9.97 $\pm$ 0.01	1.54 $\pm$ 0.00	35.44 $\pm$ 1.08
Activity Prompting	Gesture	CLIP ViT-B/32	87.5M	5.8M	87.4B	23.69 $\pm$ 0.93	10.01 $\pm$ 0.14	1.50 $\pm$ 0.00	35.97 $\pm$ 0.61
Activity Prompting	Interaction+Error	CLIP ViT-B/32	87.5M	9.1M	87.5B	22.69 $\pm$ 0.09	9.17 $\pm$ 0.01	1.54 $\pm$ 0.00	36.00 $\pm$ 1.74
Activity Prompting	Interaction	CLIP ViT-B/32	87.5M	8.0M	87.5B	23.98 $\pm$ 0.68	9.83 $\pm$ 0.18	2.34 $\pm$ 1.14	36.19 $\pm$ 1.41
Activity Kinematics Fusion	Error	CLIP ViT-B/32	87.5M	5.8M	87.4B	24.18 $\pm$ 0.50	9.55 $\pm$ 0.01	1.61 $\pm$ 0.01	37.98 $\pm$ 1.39
Activity Kinematics Fusion	Gesture+Error	CLIP ViT-B/32	87.5M	9.4M	87.5B	22.75 $\pm$ 0.56	9.91 $\pm$ 0.01	1.64 $\pm$ 0.01	37.67 $\pm$ 0.80
Activity Kinematics Fusion	Gesture	CLIP ViT-B/32	87.5M	6.1M	87.4B	22.56 $\pm$ 0.81	9.95 $\pm$ 0.01	1.59 $\pm$ 0.00	36.62 $\pm$ 0.29
Activity Kinematics Fusion	Interaction+Error	CLIP ViT-B/32	87.5M	9.4M	87.5B	22.69 $\pm$ 0.48	9.97 $\pm$ 0.10	2.27 $\pm$ 0.06	37.62 $\pm$ 1.93
Activity Kinematics Fusion	Interaction	CLIP ViT-B/32	87.5M	8.3M	87.5B	22.54 $\pm$ 1.11	9.90 $\pm$ 0.01	1.64 $\pm$ 0.01	38.18 $\pm$ 0.48
Activity-Aware Embeddings	n/a	CLIP ViT-B/32	175M	2.9M	87.4B	22.69 $\pm$ 0.42	19.78 $\pm$ 0.02	0.99 $\pm$ 0.09	45.71 $\pm$ 1.54